

# Cody Kochmann

Principal Infrastructure Engineer, Security Developer & DevSecOps Enthusiast

443-538-9639 | kochmanncody@gmail.com | <https://github.com/CodyKochmann>

## Summary

I'm a principal infrastructure engineer, security developer, and DevSecOps enthusiast. Whether you need to accelerate your developer workflows through automation, harden your infrastructure against state-sponsored attacks, or design high-performance distributed systems, I have a proven track record of driving mission-critical company goals.

## Technical Skills

- **Infrastructure & Cloud:** Kubernetes (EKS, AKS, GKE), Terraform, AWS, Azure, Linux, CloudFormation, ArgoCD, Docker
- **Security & Operations:** DevSecOps, Istio, Falco, Vault, Trivy, Threat Detection, Vulnerability Management, RBAC, Grafana, ELK/HELK Stack
- **Languages & Automation:** Python, Golang, Rust, Bash, CI/CD (GitLab, GitHub Actions, Buildkite), Infrastructure as Code (IaC)

## Experience

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### NICO - 2025 to Present

**Roles:** Lead Infrastructure Engineer

**Programming Languages:** Bash, Python, YAML

**Technologies:** Azure, Kubernetes (AKS), Terraform, ArgoCD, Argo Workflows, Grafana Stack, Istio, Falco, HashiCorp Vault (OpenBao), SealedSecrets, Zot, Trivy, Agentic AI (Claude, Hermes)

#### Achievements:

- **Security & Operations:** Hardened network traffic by deploying an Istio service mesh, configured Falco for real-time intrusion detection (IDS), and established strict RBAC across all environments.
  - **Supply Chain Security:** Secured the software supply chain by building isolated build environments, integrating Trivy for automated container vulnerability scanning, and managing the Zot registry. Safely managed credentials utilizing Vault and SealedSecrets.
  - **Cloud Infrastructure:** Led the comprehensive overhaul of the Third-Party Risk Management (TPRM) Azure infrastructure, establishing a mature, highly resilient posture rooted in industry best practices.
  - **Kubernetes & IaC:** Architected and executed the migration of all live infrastructure to Kubernetes, fully codifying the environment into modular, reusable Terraform components.
  - **AI-Accelerated Operations:** Leveraged Agentic AI tooling as a force multiplier to rapidly architect, test, and deploy infrastructure as code, significantly accelerating project delivery timelines.
  - **GitOps & Automation:** Championed the adoption of immutable infrastructure principles by implementing GitOps workflows with ArgoCD.
  - **Observability & Monitoring:** Designed and deployed centralized observability and logging pipelines using the complete Grafana stack.
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### Independent AI Researcher - 2024

**Roles:** Independent Researcher, AI Consultant

**Programming Languages:** Python, Bash

**Technologies:** Large Language Models (LLMs), Agentic Frameworks, Prompt Engineering, Vector Databases

**Achievements:**

- Designed and tested secure sandboxing methods to safely provide LLM capabilities within highly sensitive and restricted environments.
  - Analyzed the security risks and hallucination vectors of deploying LLMs in enterprise environments, developing strategies for safe adoption.
  - Assisted in critical research initiatives focused on optimizing AI operational spend and reducing infrastructure overhead.
  - Researched and implemented local GPU optimization techniques to maximize the performance of on-premise inferencing.
  - Mastered advanced prompting techniques and agentic workflows (utilizing tools like AntigraVity, Claude and Ollama) which directly translated into immense productivity multipliers for future infrastructure automation roles.
  - Conducted extensive independent and collaborative research into Large Language Model (LLM) design, training methodologies, and practical utilization for software engineering.
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## Strake - September 2023 to December 2023

*Strake closed business in December 2023.*

**Roles:** Principal Infrastructure Engineer

**Programming Languages:** Make, Python, Bash

**Technologies:** GitHub, CloudFormation, CodeDeploy, BuildKite, Postgres, SQLite3, SQS, Lambda

**Achievements:**

- Drove the development of the tools and workflows needed to generate ephemeral development environments for engineers to safely perform isolated experiments.
  - Designed and implemented tools that provided deep insights to every corner of the company's infrastructure, development patterns, and communication effectiveness.
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## QOMPLX - September 2019 to July 2023

*QOMPLX closed business in July 2023.*

**Roles:** Principal Cloud Platform Automation Engineer, Project Lead, Kubernetes SME, Solutions Architect, Security Architect, Security Developer, Mentor

**Programming Languages:** Python, Golang, Rust, Make, Bash, Jinja, YAML

**Technologies:** GitLab, CI/CD Pipelines, CI Runners, Kubernetes, Docker, Terraform, Linux Kernel, rsyslog, syslog-ng, Postgres, SQLite3, perf, eBPF, LVM2, Loki, Grafana, Kafka, Elasticsearch, Logstash, Kibana, Prometheus, Aqua, Sysdig, Twistlock, CrowdStrike, Linux Kernel, Digital Ocean, Linode, AWS (EKS, ECS, EC2, Lambda, CloudTrail, SQS, S3, Lambda, Config, IAM), Istio, Argo, Debian, Amazon Linux, CoreOS, Ubuntu, Fedora, FreeBSD, CentOS, VMWare

**Achievements:**

- **Security & Threat Management:** Drove security and vulnerability scanning operations company-wide. Identified and prevented active security breaches and remediated thousands of legacy infrastructure vulnerabilities.
- **Infrastructure & Kubernetes:** Spearheaded the company's comprehensive migration to Kubernetes as a Subject Matter Expert (SME) and Security Architect.
- **Client Operations:** Salvaged major client relationships by directly resolving complex product issues in live production environments and managing the security of specialized on-prem deployments.

- **Automation & CI/CD:** Led the design and development of the company's GitLab organization and CI/CD pipelines, building tools that enabled development teams to securely automate their own deployments.
- **Observability:** Architected centralized log collection systems, providing unprecedented levels of observability across all operational layers of the company.
- **Technical Leadership:** Acted as the platform team's Scrum Master and held weekly lectures for developers on defensive coding, infrastructure optimization, and high-performance architecture.

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## EMBEDDED FLIGHT SYSTEMS INC (NASA) - January 2017 to July 2019

**Roles:** Network Operations Engineer, Security Developer, Mentor

**Programming Languages:** Python, Golang, Rust, Bash

**Technologies:** Docker, Kubernetes, rocket, Twistlock, Sysdig, Aqua, HELK, Security Onion, OpenShift, Kubernetes, Docker, Amazon Lambda, Amazon ECS, Amazon EKS, Digital Ocean, Google GKE, Systemd, RPM, Debian, Amazon Linux, CoreOS, Ubuntu, Fedora, FreeBSD, CentOS, Cisco, VMWare

### Achievements:

- **Security Architecture:** Designed and built a high-performance DDoS mitigation tool capable of protecting NASA's network from state-sponsored attacks.
- **Security Operations:** Architected a high-performance distributed ELK cluster to ingest and analyze network logs across our hybrid cloud / on-prem network. Designed and implemented a multitude of netflow/packet capture/firewall log analysis tools to give the company deeper insight to AWS, on-prem, and container network traffic.
- Performed security analysis for numerous black box solutions to identify how safe the tools were to use and how we could keep them locked down when risks were identified.
- Led the efforts for analyzing which container security analysis tools were worthy of NASA's security standards.
- Taught developers about the principles of defensive coding and how to build highly resilient architectures that met NASA's reliability and security standards.

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## Open Source Development | June 2013 - Present

I am the lead author and architect for all of the following projects.

| Project          | Description                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------|
| battle_tested    | Fully automated Python fuzzer that highlights every crash a piece of code will raise in production. |
| generators       | High-performance pipeline processing library written in pure Python.                                |
| graphdb          | The fastest pure Python graph database available on PyPI.                                           |
| queued           | Turns functions into fully functional multi-core queued services.                                   |
| strict_functions | Function decorators for more secure Python code execution.                                          |
| stricttuple      | Python tuples with Rust-style built-in rule enforcement.                                            |
| litecollections  | Python collection types except they're all backed by SQLite.                                        |

## References

All of the following were solid people I had the honor to work with and would love to work with again one day.

| Name                | Title & Company                                                 | Contact                                                                                   |
|---------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Marcin Pohl         | Systems / Security Engineer @<br>NASA                           | marcin.pohl@nasa.gov                                                                      |
| Adam Younce         | CTO @ Strake                                                    | ayounce@ripcord.net                                                                       |
| Beck Norris         | Information Security / Compliance<br>Expert @ Frontier Airlines | rrnorris33@gmail.com                                                                      |
| Mason Walton        | Automation Engineer @ QOMPLX                                    | fin.mwalto7@gmail.com                                                                     |
| Jason Hurley        | Platform Team Lead @ QOMPLX                                     | j.m.hurley@gmail.com                                                                      |
| Nic Ladas           | Senior Platform Architect @<br>QOMPLX                           | <a href="https://www.linkedin.com/in/nicladas/">https://www.linkedin.com/in/nicladas/</a> |
| Christopher Mishaga | CISO @ NASA                                                     | christopher.a.mishaga@nasa.gov                                                            |